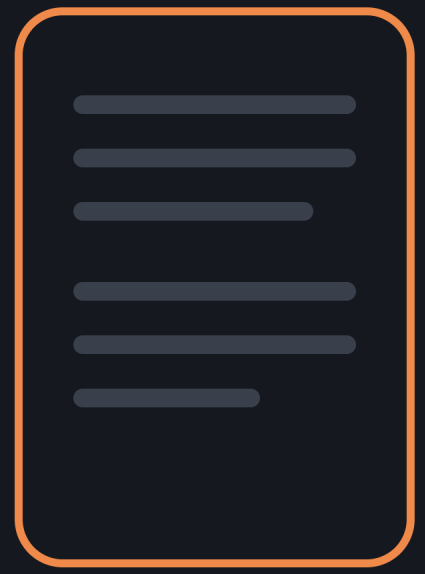
OPEN SOURCE · MACOS + IPADOS

PortraitDisplay

Your iPad as a portrait second screen for your Mac, over USB-C, Wi-Fi or Safari.



github.com/ggmax-gif/PortraitDisplay · MIT licence · Daniel Fung, 2026

WHY IT EXISTS

Sidecar wouldn’t do portrait

On my MacBook, Sidecar only shows the iPad in landscape. For reading documents I wanted a tall page.

Streaming BetterDisplay to Sidecar only showed black, so I built my own: a portrait monitor that exists only in software, streamed to the iPad.



Sidecar: landscape

Wanted: portrait

WHAT IT DOES

Two ways to view it

	Safari	iPad app
Install	Nothing: open a link	Xcode build (7 days if free)
Connection	Wi-Fi via the router	USB-C, or direct Wi-Fi
Link round trip	12.7 ms median	0.31 ms USB, 5.2 ms direct
Input	Tap, drag, scroll, right-click	+ pinch, rotate and swipes
Rotation	Portrait only	Turns with the iPad

Link round trips: echo test over each path, 27 Oct 2026 (Spike/link/ in the repo).

WHAT IT DOES

Features

Built for portrait

A tall page for reading. Turn the iPad and the Mac’s display turns with it.

USB-C, about 6 ms

Frames reach the iPad and come back in about 6 ms over the cable. No network needed.

Direct Wi-Fi

Unplugged, it goes peer to peer, no router, over TLS with a key paired by cable.

Touch and Pencil

Pencil pressure and tilt, taps, scrolling, pinch, rotate and three-finger swipes.

Sharp Retina text

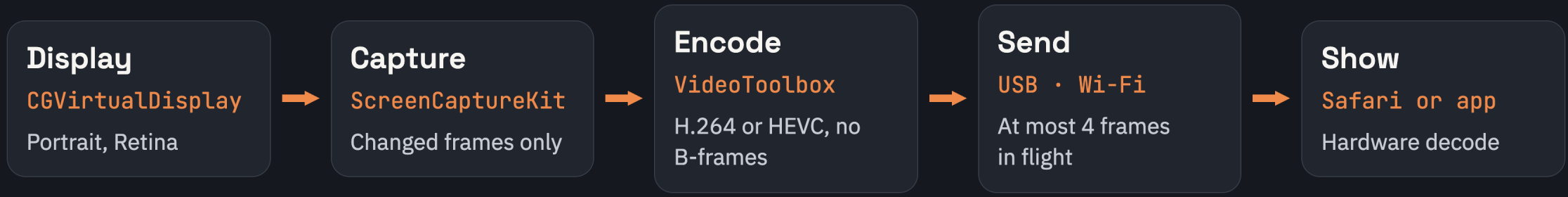
2048×2732 px on a 12.9-inch iPad, sized to match it, in hardware H.264 or HEVC.

Or just Safari

No app at all: open a link in Safari and add it to the Home Screen.

HOW IT WORKS

How a frame travels



Touch, Pencil and acknowledgements come back the same way; the input injector replays them on the virtual display with CGEvent.

A menu bar app starts and stops it. With on-demand on, the default, the display exists only while an iPad is connected.

HOW IT WORKS

Measured latency

0.31 ms
USB link round trip, median (p99 0.94 ms)

5.2 ms
Direct Wi-Fi link round trip, median

12.7 ms
Link round trip through the router, median

12 ms
Hardware encode of one 2048×2732 frame

In use, a frame’s round trip is about 6 ms over USB and 7 to 10 ms over direct Wi-Fi, with spikes to about 200 ms while the Mac’s radio also serves the router.

Link test: echo over each path, 27 Oct 2026 (Spike/link/). In-use figures: the host’s stats line, 27–28 Oct 2026.

HOW IT WORKS

Security

Paired by cable

Over USB, the Mac gives the app a 256-bit key. Direct Wi-Fi uses TLS with that key, so a device without it can’t connect or listen.

Private IPs only

Safari needs HTTPS, so the Mac runs its own certificate authority. It is limited to private network addresses and can’t vouch for real websites.

A key in every link

The Safari page only streams to a link carrying its key, so other people on the same network can’t watch your screen.

No server in between: the stream stays between your Mac and your iPad.

HOW IT WORKS

How it’s built

4,919
Lines of Swift, Mac and iPad

0
Third-party packages

42
Unit tests, run by CI on every push

1
Private API: CGVirtualDisplay

Tests cover message framing, gesture decisions, the direct Wi-Fi TLS handshake over loopback, and the H.264 parameter-set patch, checked against a separate SPS parser.

The host prints capture, encode, bitrate and round-trip figures every 2 seconds, and the design notes record each latency test and codec comparison.

WHERE IT STANDS

Status

Works today

- USB, direct Wi-Fi and Safari
- Rotation, auto-size and pairing with several Macs
- Tested on an Apple silicon MacBook and a 12.9-inch iPad Pro (4th generation), both on version 27.2

Limits

- The iPad app needs Xcode and an Apple ID; free installs last 7 days
- A private API keeps the Mac app off the Mac App Store
- Safari stays portrait

Next: friends try it on other Macs and iPads. Putting the iPad app on the App Store needs Apple’s paid developer programme, USD 99 a year.

Try it, or read the code

github.com/ggmax-gif/PortraitDisplay

MIT licence. Bug reports and pull requests are welcome as GitHub issues.